

It must be modified depending on the type of analysis:

0: sample from prior

1: clock model selection

2: use the gradient and Hessian

3: to approximate likelihood (gradient and Hessian)

Evolutionary
model

```
seed = -1          * Seed number, type -1 to generate a random seed number
seqfile = ../concatenated.phylip * Path to input sequence file (e.g., many alignments in a unique file)
treefile = ../startingTree.phy  * Path to input tree file including calibration points as node labels
outfile = out.log    * Path to log output file
mcmcfile = mcmc.txt  * Path to output file with MCMC samples

ndata = 5           * Number of partitions
seqtype = 0         * 0: nucleotides; 1:codons; 2:AAAs
usedata = 1         * 0: no data (samp from prior); 1:seq likelihood; 2:approx likelihood (use in.BV); 3:out.BV
clock = 1           * 1: global clock; 2: independent rates; 3: correlated rates

model = 4           * 0:JC69, 1:K80, 2:F81, 3:F84, 4:HKY85. Pick 4 for nt, and 2 for aa
alpha = 0.5         * alpha for gamma rates at sites
ncatG = 5           * No. categories in discrete gamma

cleandata = 0       * remove sites with ambiguity data (1:yes, 0:no)?
```

priors
birth-death
process

```
BDparas = 1 1 0.1  * birth, death, sampling
kappa_gamma = 2 .2  * gamma prior for kappa
alpha_gamma = 2 4   * gamma prior for alpha
rgene_gamma = 2 20  * gamma prior for overall rates for genes
sigma2_gamma = 1 10 * gamma prior for sigma^2 (for clock=2 or 3)
```

MCMC
sampling

```
print = 1           * avoid printing branch rates for now as the tree is very large
checkpoint = 1       * turn checkpointing on with 1, change to 2 if resume is required after a failed one where 1
burnin = 4000        * Number of iterations that will be discarded as part of burn-in
sampfreq = 6         * Sampling frequency
nsample = 20000       * Number of samples to be collected
```

* burnin, samplefreq, and nsample imply that a total burnin + (sampfreq * nsample) samples will be collected with mcmc